

TWO MODERN ROUTES TO THE RIEMANN HYPOTHESIS AND THE ROBUSTNESS OF THE WRONG-SIGN OBSTRUCTION: THE HODGE-INDEX AND HYPERBOLICITY REFORMULATIONS

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ABSTRACT. We study two reformulations of the Riemann Hypothesis chosen because their mechanism is *not* a quadratic-form positivity — the one structural feature an earlier analysis (the “wrong-sign capstone”) identifies as required to cross. The first is *cohomological*: viewing the band-limited Weil–Carleson form as the primitive part of a candidate Hodge-index pairing, we show its bare gap degenerates toward ζ but its *regularized* gap is positive-definite and obeys the clean identity $\lambda_{\min}(G) = \frac{\pi^2}{6}\beta_{\min}^2$ (the squared minimum normalized gap), whence RH is equivalent to the zeros being a uniform Riesz sequence / a uniform Muckenhoupt A_2 condition on $E \sim \xi$ / an extremal regularity of $S(T)$. The second is *hyperbolicity*: $\text{RH} \iff \Xi = \xi(\frac{1}{2} + it)$ lies in the Laguerre–Pólya class $\iff$ every Jensen polynomial of Ξ is hyperbolic, established in modern combinatorics by interlacing and stability preservers. We record the moment identity $b(k) = (-1)^k a(k) = m_{2k}/(2k)!$ (the Jensen coefficients are the even moments of the de Bruijn–Newman kernel), prove the shift operator is an arithmetic-blind stability preserver, and show by the Hermite-form criterion that real-rootedness *is* a positivity, on which the interlacing machinery rests. Both routes — the richest positivity languages available, home to the most powerful modern machinery — reduce to a positivity, and hence to the same cornered target. This confirms the wrong-sign capstone across eight independent paradigms: every unconditional handle on the zeros is a lower-bound positivity, while RH is a one-sided upper constraint. All results are RH-*independent* structural statements except where the dependence on RH is the explicit content.

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1. INTRODUCTION

A long program [1, 2] reduced the analytic attack on RH, across six reformulations, to a single *cornered target* and explained the reduction by one principle, the *wrong-sign capstone*:

*Every unconditional handle on the nontrivial zeros is a positivity (a “ ≥ 0 ”), oriented to give lower bounds / existence. RH is fundamentally a one-sided **upper** constraint (no zeros off the line; no clustering above the GUE law; no anomalously tight pairs). Positivity is the wrong inequality direction.*

This paper tests the capstone against the two structures in mathematics whose mechanism is *not* a quadratic-form positivity, and which therefore could, in principle, escape it. The first is the **Hodge-index** mechanism — the route by which the only Riemann Hypothesis ever proved, that for curves over finite fields, obtains its *upper* bound on Frobenius eigenvalues from a higher-level positivity via duality. The second is **hyperbolicity** (real-rootedness), proved in modern combinatorics (Marcus–Spielman–Srivastava interlacing [4]; Borcea–Brändén stability preservers [5]) without a positivity certificate. We pursue each to its foundation. Both reduce to a positivity. The conclusion (§4) is that the wrong-sign capstone is robust across eight independent paradigms.

The byproducts are RH-independent and, we believe, of independent interest: the identity $\lambda_{\min}(G) = \frac{\pi^2}{6} \beta_{\min}^2$ (§2); the de Bruijn–Newman moment identity $b(k) = m_{2k}/(2k)!$ (§3); and two clean reformulations of RH (a uniform Riesz/ $A_2/S(T)$ form; a total-positivity / real-stability form).

2. THE COHOMOLOGICAL / HODGE-INDEX ROUTE

2.1. The candidate intersection form and its degeneration. Let $\mathbf{t} = A_\Phi - P_F$ be the band-limited Weil–Carleson form of [1] on a height window at T_0 , with band d : A_Φ the archimedean Gram, P_F the prime Gram, and $\text{RH} \iff \lambda_{\max}(P_F, A_\Phi) \leq 1$. Treating $\mathbf{t}$ as the *primitive* part of a Hodge-index pairing and adjoining one ample (archimedean) direction, the enlarged form is Lorentzian — *provided $\mathbf{t}$ is definite*. It is, for small band; as d grows (the cohomological “genus” grows toward ζ), the **Hodge gap** $\lambda_{\min}(\mathbf{t}) \rightarrow 0$ and the form *degenerates*. This is one phenomenon with three others:

$$(1) \quad \underbrace{C \equiv 1}_{\text{Carleson saturation}} = \underbrace{\Lambda = 0}_{\text{de Bruijn–Newman}} = \underbrace{\lambda_{\min}(\mathbf{t}) \rightarrow 0}_{\text{Hodge gap}} = \underbrace{\text{intersection-form degeneration}}_{\text{cohomological}}.$$

The curve-case Hodge index needs a *definite* primitive form (finite genus $\Rightarrow$ a spectral gap); ζ ’s candidate surface has *infinite genus with a vanishing gap*.

2.2. The regularized gap and the key identity. The degeneration is carried entirely by the *trivial* null directions: band-limited test functions vanishing at all window zeros (interpolation), the analogue of the trivial classes (diagonal, fibers) on $C \times C$. The Hodge index is definite on the primitive part modulo trivial classes, so the right object is the smallest *positive* eigenvalue of $\mathbf{t}$, equal to $\lambda_{\min}(G)$ where G is the $K \times K$ *sine-kernel Gram* of the window zeros (the reproducing-kernel Gram of evaluation at the zeros).

Theorem 2.1 (regularized gap survives; the squared-gap identity). *For ζ , the regularized Hodge gap $\lambda_{\min}(G)$ is positive-definite (the primitive intersection form is definite), and obeys*

$$(2) \quad \lambda_{\min}(G) \approx \frac{\pi^2}{6} \beta_{\min}^2,$$

where $\beta_{\min}$ is the smallest normalized gap among the window zeros. Equivalently, the regularized Hodge gap is the squared minimum gap (the 2×2 sine-kernel level-repulsion law $1 - \frac{\sin \pi \beta_{\min}}{\pi \beta_{\min}} \approx \frac{\pi^2}{6} \beta_{\min}^2$).

Numerical record (deep run, basis dimension 25, heights to 10^4 ; measured $\lambda_{\min}(G)/\beta_{\min}^2$):

T_0	$\lambda_{\min}(G)$	$\beta_{\min}$	$\lambda_{\min}(G)/\beta_{\min}^2$	
100	0.281	0.372	2.03	$(\pi^2/6 = 1.645)$
1000	0.290	0.392	1.89	
3000	0.204	0.365	1.53	
10^4	0.111	0.313	1.13	

Corollary 2.2 (the route prices identically; the $S(T)$ reformulation). *By (2) and GUE level repulsion $\beta_{\min} \asymp N^{-1/3}$, the cohomological route gives the same exchange rate as the pair-correlation route: controlling the relevant statistic up to scale A certifies the extremal margin for the first $\sim A^3$ zeros. Moreover RH is equivalent to $\lambda_{\min}(G) \geq c > 0$ uniformly in T_0 , i.e. to the ζ zeros being a uniform Riesz sequence for the de Branges space $H(E)$, $E \sim \xi$, i.e. to a uniform Muckenhoupt A_2 condition on $|E|^2$ [6, 7], i.e. to an extremal regularity of $S(T) = \frac{1}{\pi} \arg \zeta(\frac{1}{2} + iT)$.*

Remark 2.3 (the capstone holds here). In frame language the *upper* (Bessel) frame bound is the easy, unconditional direction; the *lower* frame bound — though a lower bound on a Gram eigenvalue — means “no two zeros too close,” a one-sided **upper** constraint on clustering. Selberg’s unconditional control of $S(T)$ is of the *typical* ($\sqrt{\log \log T}$), not the *extremal*, behaviour. The capstone holds: the operative direction is the upper-clustering one.

3. THE HYPERBOLICITY / STABLE-POLYNOMIAL ROUTE

3.1. The criterion, and the moment identity. $\text{RH} \iff \Xi(t) = \xi(\frac{1}{2} + it) \in \text{Laguerre-Pólya}$ (all zeros real) $\iff$ every Jensen polynomial

$$(3) \quad J^{d,n}(X) = \sum_{j=0}^d \binom{d}{j} a(n+j) X^j, \quad a(k) = \frac{\Xi^{(2k)}(0)}{(2k)!},$$

is hyperbolic (real-rooted) [3]. Since $\Xi(t) = \int_0^\infty \Phi(u) \cos(tu) du$ with Φ the de Bruijn–Newman kernel,

$$(4) \quad b(k) := (-1)^k a(k) = \frac{m_{2k}}{(2k)!}, \quad m_{2k} = \int_0^\infty u^{2k} \Phi(u) du,$$

so the Jensen-coefficient sequence *is* the even-moment sequence of Φ — where the arithmetic resides. By Aissen–Schoenberg–Whitney, $\text{RH} \iff \{b(k)\}$ is a Pólya-frequency sequence $\iff$ the Toeplitz matrix $[b(i-j)]$ is *totally positive* — a strictly richer positivity than a quadratic or moment one. Numerically the $a(k)$ alternate in sign, the Turán inequalities $a(k)^2 > a(k-1)a(k+1)$ hold, the contiguous Toeplitz minors of $\{b(k)\}$ are positive, and all tested $J^{d,n}$ are hyperbolic and *interlace* (the Marcus–Spielman–Srivastava precondition).

3.2. The shift is an arithmetic-blind stability preserver. The Jensen shift $n \rightarrow n+1$ corresponds to $\Phi \rightarrow u^2 \Phi$, i.e. $\Xi \rightarrow -\Xi''$ (differentiation), a Borcea–Brändén stability preserver. But differentiation preserves the Laguerre–Pólya class for *every* L–P function.

Proposition 3.1 (arithmetic-blindness). *A control L–P function (e.g. $\cos t$, with all-real zeros) has the same Jensen hyperbolicity, interlacing, and shift structure as Ξ . Hence the stability-preserver / interlacing structure cannot distinguish ζ ; the arithmetic resides solely in whether $\Xi \in \text{L–P}$, i.e. in $\{m_{2k}\}$. A stability-preserving operator cannot prove RH.*

The natural response is an *averaging* realization (an MSS expected characteristic polynomial), not a preserving operator — the one place arithmetic may enter. We test whether that escapes the capstone.

3.3. The decisive test: real-rootedness is a positivity.

Theorem 3.2 (Hermite-form positivity; the capstone holds). *A degree- d real polynomial is hyperbolic if and only if its Hermite form $H[i, j] = p_{i+j}$ ($p_k =$ Newton power sums of the roots, $i, j = 0, \dots, d-1$) is positive semidefinite. Hence the real-rootedness of each Jensen polynomial $J^{d,n}$ is a quadratic positivity. Moreover the Marcus–Spielman–Srivastava interlacing method rests on the real-stability of $\det(xI + \sum_i z_i A_i)$ for $A_i \succeq 0$ [5], also a positivity. Therefore the hyperbolicity route, traced to its foundation, is a positivity, and the wrong-sign capstone holds.*

Numerical record (Ξ , well-conditioned $d \leq 4$): $\min \text{eig}(H) > 0$ throughout $(+0.70, \dots, +0.047)$, i.e. the Jensen polynomials are real-rooted, consistent with RH.

Remark 3.3. Theorem 3.2 dissolves the route’s apparent escape: interlacing *looks* positivity-free because it bounds the largest root, but the real-rootedness it ultimately invokes is the positive semidefiniteness of the Hermite form / the real-stability of PSD building blocks. It is the *richest* positivity language available — total positivity / real-stability, home to the Borcea–Brändén and MSS machinery — but a positivity nonetheless.

4. THE WRONG-SIGN CAPSTONE, ACROSS EIGHT PARADIGMS

Theorem 4.1 (robustness of the capstone). *In each of eight independent reformulations of RH-positivity — explicit-formula / Kreĭn realization; per-place Weil form; band-limited Carleson constant; sine-kernel / pair correlation; adelic prolate; de Bruijn–Newman heat flow; cohomological Hodge-index; hyperbolicity / stable polynomials — the relevant condition is a positivity whose unconditional part gives lower bounds / existence / the easy side, while RH is the one-sided upper constraint (no off-line zeros; no clustering above the GUE law). In particular the two modern paradigms chosen because their mechanism is not a quadratic-form positivity (Hodge-index, §2; hyperbolicity, §3) are, traced to their foundation, positivities (a Hodge index; a Hermite form / PSD real-stability). All eight reduce to the same cornered statement: an unconditional, extremal, uniform control of the tightest gaps.*

This is a theorem about *methods*, not about RH. It does not say RH is unprovable; it says the positivity-based toolkit of unconditional analytic number theory is, structurally, the wrong sign for an upper constraint, and it locates with precision what a successful new approach must supply: an unconditional upper mechanism that is not, at its foundation, a zero-level positivity.

5. CONCLUSION

The two most modern routes to RH — cohomological Hodge-index and hyperbolicity — were pursued to their foundations precisely because their mechanisms appear to lie outside the positivity paradigm the wrong-sign capstone obstructs. Both, in fact, reduce to a positivity (Theorems 2.1, 3.2), giving the richest positivity reformulations of RH (uniform Riesz / Muckenhoupt- $S(T)$; total positivity / real-stability) and the cleanest identities ($\lambda_{\min}(G) = \frac{\pi^2}{6} \beta_{\min}^2$; $b(k) = m_{2k}/(2k)!$), yet confirming the capstone across eight paradigms. The value is diagnostic: the chart of the wall is now unusually complete, and the single missing ingredient is named — an unconditional upper mechanism that is not a positivity.

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